

Design and Technology NEA

Aarav Thakur

10XO

Mr Laher

Problem 1

Delicate Plates

Since the majority of plates are made of glass, porcelain or ceramic, they make a highly unpleasant sound when scratched with a fork, they easily break and when they break, they shatter meaning that it is difficult to clean up all of the shards of plate and they shards are very dangerous as accidentally stepping on them could severely injure the person

A study by the NIH focusing on the injuries of minors has shown that 86% of lacerations are caused by minors stepped on shards of glass in 2015.



One solution to this problem is tempered glass which is a form of glass which has been heat treated to become stronger than regular glass. One problem with this solution is that the plate can still break on impact making it not ideal for a plate



To solve this problem, I could make a plate which is hard, does not shatter and has an appropriate design and still maintain a premium feel.



Potential Client: My mother

Potential Target Market: Restaurants

Problem 2

Heavy Guitars

The average weight of a guitar is around 3.6 kg and is almost 1 metre long this makes it difficult for people including many of my guitarist friends to practice in different places. Although their passion for guitar pushes them to practice whenever they can, the heavy weight and large size of the guitar makes it difficult to do so. In addition, electric guitars require a large amount of gear to get the correct sound that one looks for and often times, this gear can be expensive and heavy further restricting people from practicing in places other than their home.

As shown in the table: Throughout the majority of the current decade, guitar sales have been on the rise and have increased by 6.4% in the past year alone according to Fender's 2024 sales report. This suggests a recovering market.



Ways I could help: When designing the guitar I must take into account everything such as but not limited to tone, feel and comfort while still optimising its smaller size and lightweight.



Potential Clients: Shiv, Ubai, Kenda, Rocco

Potential Target Market: Guitarists

Problem 3

Robotics Education

Although movement such as FIRST have revolutionized robotics among high schoolers, organising events and competitions all around the world encouraging more teenagers to engage in robotics, it has a long way to go before reaching all students in all schools. For example, in countries like Portugal, Spain and Belarus, there are no initiatives which encourage robotics. This has become a hindrance to many people who are actually interested in the subject but it is either too expensive or too distant to be worthwhile



- By 2030, there could be a **global shortage of over 85 million skilled workers**, many in STEM fields. **Source:** Korn Ferry – *The Global Talent Crunch report 2023*



- Only about 40% of schools worldwide** have access to adequate STEM learning resources. **Source:** UNESCO – global STEM education reports 2024



Ways I could help:

I could create a cheap robotics kit which can be easily distributed to schools all around the world and teach them about engineering and robotics.

Potential Client: Faris Tarraf

Potential Target Market: Robotics enthusiasts and young children

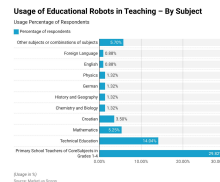
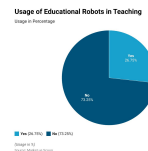
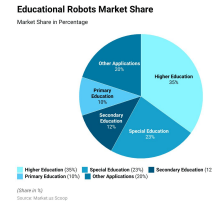
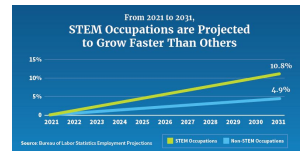
In-depth research

Why I chose problem 3:
Due to my background in robotics, I believe this is more suitable for my skill set and enables me to take into account more factors that I wouldn't have been able to if I had picked the other 2. Furthermore, a robotics kit is much more niche and marketable compared to a guitar in which the big companies have already been established. For this reason, it would be difficult for a small company to thrive in this economic environment despite my musical background in Robotics. Additionally, I have no experience in pottery or any related field so I would most likely struggle and it would result in a less than acceptable product.

Benefits of robotics education: Robotics education is education involving the interdisciplinary application of mechanical, electrical and software engineering. However, this process of creation is not easy and only ever becomes more challenging as one advances more into the field. This difficulty comes with a silver lining: it may build perseverance within the user. Furthermore, the difficulties of building a robot often require creative workarounds to solve problems and fosters a growth mindset by forcing them through the iterative design process. This exercises the individual's problem solving skills as described in research by Ouyang and Xu in 2024. This also helps the teenager connect disciplines together. Usually, subjects are taught separately with little to no connection between them. Robotics forces the student to connect these fields together to solve problems. These built connection have been shown to improve learning outcomes. As STEM job occupations are projected rise by 10.8% in the US alone over from 2021 to 2031 (2022 D.O.L STEM day), STEM professionals are becoming more important now more than ever. All of these studies compound together to prove that robotics and STEM in general are becoming bigger than ever and that these students will require more help now which is the gap that I am trying to fill. This also means that **I will have to focus of my design connecting numerous fields rather than specialising in a single topic.**

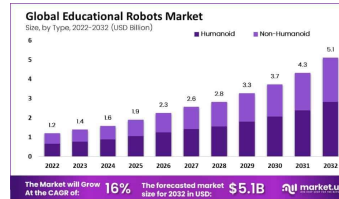
Design implications: My product must be able to provide both a basic and a complex understanding of systems and offer both structural, electrical and programming obstacles while remaining under £40 and should target middle and high school teenagers

Existing products: In 2023, the education robotics market had a revenue of USD 1.4 billion and is projected to have a USD 4.3 billion revenue by 2031 as depicted by a study conducted by [scoop.market](#). This market growth provide an opportunity for a cheaper alternative to be introduced. Furthermore, the more comprehensive a kit is, the more expensive it appears to be. As the base line is already high for robotics kits, they are usually only limited to higher educational institutions such as universities and colleges. This is reflected through their 35% market share of educational robotics as of 2026. This gap is yet to be filled by a product which can **easily teach bot beginners and advanced roboticists.**



Problems with robotics education:

General market research has depicted robotics kit at an average of USD 120 - USD 150 for advanced robotics kits such as the [Makerbot M2](#). This high cost barrier has prevented majority of students from developing their enthusiasm for robotics into a fully fledged career path. Teachers will also require training to be able to teach robotics, and this training is not vastly present in the world of today. This along with poor curriculum fit discourages teachers from integrating robotics into the classroom. This means student miss out on numerous academic advantages which would not only the students get better grades but also makes the school look more appealing. Furthermore, the advent of the internet has accelerated technological progression to astronomical amounts. Thus, maintenance along with obsolescence become regular and costly figures. To combat this, **my product will be kept cheap (no more than £40) and will be reusable in different classrooms** by applying a multitude of concepts within the design. **The documentation for the technical concepts and the application of those concepts will be included with the robot.**



Design brief

I will be designing a product that satisfies the needs and demands of middle and high school robotics enthusiasts. This product will be used over a period of 3 months to align with the duration of school terms so it must meet the requirements for that. The customer range will be between the ages of 12-18 as often they want to get started and dive deep into robotics but can only rely on small allowance money to do so. Hence, I chose an individual in this target audience. Furthermore, due to my close relationship with him, he is not afraid to tell me if a design is not aesthetically pleasing. This means I receive more critical feedback for me to develop upon.

In addition, I would like my product to be aesthetically pleasing as possible, to meet this goal I will make the product have a clean design, making it have a sleek appeal. The finish will be matte to give it a bolder appearance and to remove any extra adhesion that a glossy colour may incur (Further finish and aesthetic research on pages 19 and 7). Since teenagers often are developing their personalities, I will keep the designs simplistic.

Furthermore, the product will need to meet a range of requirements and specific needs for my client. Therefore I will need to research into my clients needs and wants through an in-depth interview and questionnaire.

The product must also be ergonomically designed and has the potential to suit multiple users, including my client and potential sellers. To do this I will do my own research on existing products, analysing the advantages and disadvantages of each product.

This product's main use will be to educate teenagers on the principles of robotics for a low price so the need for it to be functional, minimalistic and cost efficient is necessary. Hence, the product will be made of reliable and cheap materials such as PLA and plywood.

The product will also be manufactured using machinery only available within the department. Therefore any manufacturing process and materials must be used efficiently so that it minimises cost and therefore maximises profit from the sales. The materials must also be available in the department meaning that every item must be used to its maximum potential to achieve the goal of making this product.

What I plan to research	How I plan to gather the information	Why I will research it How will it help me with my project
Client information	Primary research: Interview	It will help me understand what my client wants
Aesthetics	Primary research: Interview Secondary research: Internet	It will help me design my product in an appealing way
Cost	Primary research: Interview Secondary research: Internet	It will help me set a target price on the product
Environment	Secondary research: Internet	It will help me prevent harm to the Environment
Ergonomics and anthropometric	Primary research: Client measurements	It will help me make my product comfortable to hold
Existing products	Secondary research: Internet	It will help me get ideas for my product
Size	Primary research: measurements Secondary research: Internet	It will help me set constraints on my design
Safety	Secondary research: Mr Laher and internet	It will help me prevent injury when interacting with my product
Function	Primary research: Interview Secondary research: Internet	It will help me understand what features the final robot kit should have
Materials and finishes	Primary research: Experimentation	It will help me decide which materials to use for my product
Manufacturing processes	Primary research: Experimentation through building a cube	It will help me plan how I will build my product

“Yes I have.”

It was very enjoyable

I first got into Robotics when I was young through a Lego Mindstorms Course my parents signed me up for.

The lack of proper Robotics Teams near me, as well as the language barrier.

Yes. Mechanical (I believe it would count as mechanical). A Pinewood Derby Car.

I would say 50 Euros, give or take.

I'd prefer clear instructions, instructions for individual mechanisms/features, and the ability to just figure things out myself.

Both are okay, but a clean look is nicer.

I will be using it alone

Not that I know of.

Handedness: Right



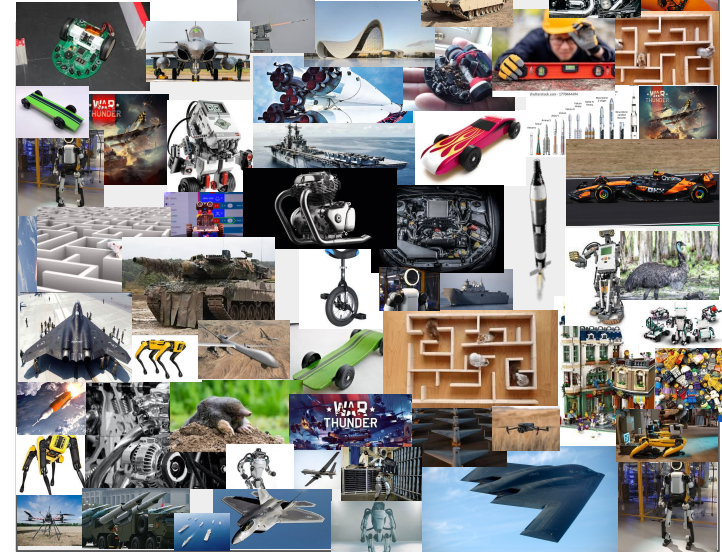
His hobbies include gaming and coding. However, due to his current relocation to portugal, he has been unable to participate in robotics. He often complains about expensive kits and the lack of programs within his area. For this reason I am making a cheap robotics kit which can be used to learn about basic engineering concepts

External sensors such as magnetometers and cameras



Since my client's work area is small, I will have to make sure that my parts do not take up much space

My target audience is children and teenagers who are interested in engineering or robotics but cannot afford to do so practically.



Aesthetics

The robotics kit on the right uses blue parts which research has shown boosts creativity (Psychology of blue by [Bojana Radovanovic](#) in 2026). This aids in the user's engagement and boost conceptualisation. Additionally, the continuous treads and the metallic coating makes the robot feel more like a tank, further interesting children and teenagers,



The use of the ultrasound sensors as eyes makes the robot appear more human forming a stronger connection between the user and the robot. Furthermore, .

Colour Psychology: As referenced in the first aesthetic analysis of a product, certain colours affect the user to express certain emotions. For example, blue is commonly associated with technology and engineering products (Psychology of blue by [Bojana Radovanovic](#) in 2026). Hence, I will use blue as a main colour and red as accents;

Biomimicry: Biomimicry is the use of the natural world to solve problems that we ourselves have. Incorporating the natural world into my product, it will teach the user that a great source of inspiration for solving real world problems is to look at nature as it often has already solved the same problem but in a different context. For example, the bullet train often made large booms whenever passing through tunnel. So engineers took to the kingfisher bird for inspiration which ultimately helped with the final design. We can apply the same concept to this kit. The "animal" that most people interact with are human so making a humanoid robot would build a greater connection between the robot and the user.

Cost



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\$3⁰⁰
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\$5.19 delivery **Mar 18 - 27**
Ages: 3 months and up

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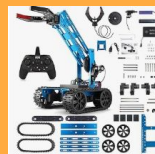


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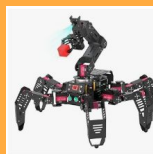


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\$10⁹⁹ Typical: \$59.99
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Teaches
mechanical and
IK concepts



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\$2,059⁹⁹
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HIWONDER Robot Car with
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ROS1 Education Lidar SLAM
Mapping Navigation AI Vision...

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Teaches mechanical
and IK concepts

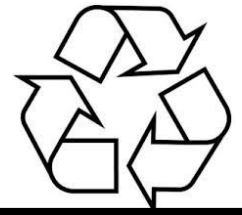
Cost of my product: The cost of the product will be determined materials I use, the current price of similar products on the market, and the time and labour that will go into making the product. Due to one of my product targets is being cheap, I will have to price the kit in the middle to low price range. The BOM on the right is an approximate of what the material costs will be. The prices listed may not be the same as what is listed on the website as prices of these components as of February 2026 are volatile.

Part	Model	Price
Micro controller	Arduino Uno	1.75
IMU	MPU 6050	1.21
DC Motor, Wheel	Kit	0.98
Battery holder	6 x AA Battery Case Shell	0.49
Servo Motors	MG995	15
Chassis	3D printed and laser cut	2
Fasteners	Zip ties	2.2
Wires	Dupont	1.89
Motor Driver,	LN298N	0.27
Total		25.79

Design implications: Since the estimated cost of materials is USD 26/ £20 (conversion is \$1 = £0.74 at time of research). Including shipping, packaging, print time and paint, it would cost closer to £25. I will price my product at USD 50/ £35 - which is severely cheaper than my competitors such as [BEV](#) who is priced at USD 695 (£3700 and £512) respectively while still teaching the almost all of robotics concepts.

Design implications: From this research, I have decided to use matte and vibrant colour for my product as it seems to be backed by both scientific and market research to maximise engagement within the target audience. I have also decided to design my ideas around with real animals or real icons/ staples of pop culture

Environment and sustainability



Fossil fuels are fuels which were formed over millions of years from dead animals and plants. They often use combustion to release energy. This releases toxic gases such as sulphur dioxide and carbon dioxide into the atmosphere and contribute to global warming. Nuclear energy is energy derived from the nuclear decay of enriched uranium 235. This produces radioactive waste which is difficult to dispose of and takes centuries to decompose. Renewable energy is energy derived from renewable sources such as wind and the sun but is unpredictable and only produces energy in intervals. A finite resource is a resource which will eventually run out. An infinite resource is a resource which is able to be renewed at a rate in which the new resource can replace the old resource. Though reliable versions of this does not exist, we should preserve the energy we do have. For this reason, I have decided to use an easy-to-manufacture material such as PLA which does not require much energy to deform but remains still at room temperature.

Materials that can be recycled:

- 1) Plastics*
- 2) Cardboard
- 3) Wood
- 4) Metal

Materials which cannot be recycled:

- 1) Styrofoam
- 2) Ceramic
- Teflon

* Only certain forms of this material can be recycled and only in specialised facilities

The recycling symbol three arrows symbolize the process of collecting, manufacturing, and purchasing products made from recycled materials. Recycling is needed to conserve natural resources like trees and minerals by using materials which were previously used to make products; reduce energy consumption and greenhouse gas emissions, decrease pollution from landfills and waste, protect ecosystems and wildlife.

LCA is a method to determine a product's impact on the environment. LCA looks at the entire life cycle of the product to prevent one impact from being shifted to another. The LCA is a standardised industrial practice which is governed by ISO 14040 and 14044.

What are the 6Rs:

- 1) Rethink: think about using non-recyclable materials before using them (careful consideration of materials)
- 2) Refuse: try your best to avoid using non-recyclable materials such as styrofoam
- 3) Reduce: limit the amount of non-recyclable materials used (optimize infill setting to limit plastic use).
- 4) Reuse: limit the amount of materials you waste by using scrap material which otherwise would have been considered as trash (use tools to resize parts without having to print them again)
- 5) Repair: try fixing things before throwing them out
- 6) Recycle: if it is not possible to reuse the materials, try recycling the materials. (Throw away all plastic into a recycling bin)

Design implications: From this research, I must use materials that are biodegradable and will have to follow ISO 14040 and ISO 14044. Hence, I am inclined to use woods, PLA and possibly metals such as aluminium. Since PLA is often recycled and can be melted down back into filament, I do not need to be overly cautious with my usage of the material during the ideation and prototyping stage. In practice, I will document the material sources and disposal route for each component. However, the ease of use of styrofoam does make it enticing for initial prototyping but the environmental harm it will cause has pushed me away from using it.



Aesthetics:
The robot is mainly black with yellow and blue accents. This along with its barebones nature makes the product look aesthetically pleasing while remaining functional

Size:
The robot is a compact robot car for a tabletop. Hence can be used on desks and is easy to store and transport.

Cost:
The cost of the product is around USD 100. However, this is only the cost within the US. I believe this is a fair price as it offers high quality parts such as acrylic base plates, arduinos, servos etc.

Customer:
The product has clearly been designed for a customer to build themselves as it include not only the parts that are needed but also the physical tools that would be used. **Following this I will include a screwdriver (developed in design implications on page 11)**

Environment:
The product is mostly acrylic and electronics. Hence, the product itself is not environmentally safe. However, the parts could be reused in separate projects making the product less environmentally threatening. However, the acrylic will have to be thrown away once used as it cannot be used for be used for any other project.

Product Analysis



DR CLINT

★★★★★ Wonderful product as a STEM kit!!!

Reviewed in the United States on February 6, 2026

Verified Purchase

This Smart Robot was my introduction to Elegoo products and the experience has made me a diehard fan of Elegoo. I bought this for my 9 year old son as a gift for the holidays. This was my attempt to give him gifts would peak his interest enough to get him away from all the games and electric devices and this did the trick. At first when I first opened it I wondered if it was too much for him to assemble but boy was I wrong. He enjoyed every moment of it and has not stopped using it. Its so much fun that everyone in the house is enjoying it. It was designed and packaged so well and the instructions were very easy to follow. Love the care they took and the packaging was neat and well organized. In its default state its packed with features but the intro to Arduino IDE increases its value immensely making it possible to add more components, features and functionality. This has already paid for itself times over and its value for money is undeniable!!



Happy Shopper

★★★★☆ SO-SO

Reviewed in the United States on December 28, 2025

Verified Purchase

The good: In spite of it seeming like a toy, it is a very complex electro-mechanical device. The kit is assembled with well illustrated instructions.

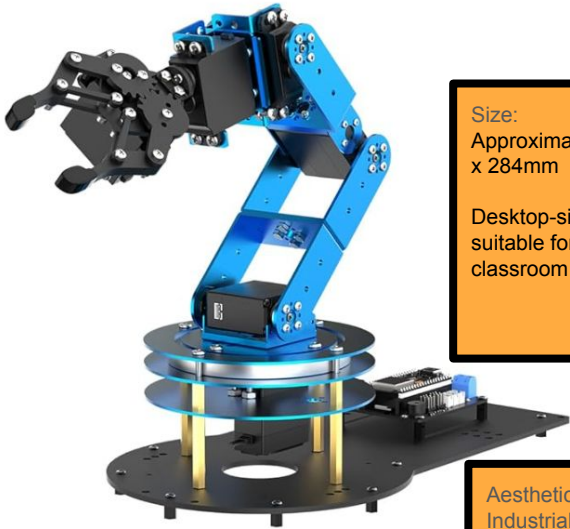
The so so: I don't think the average 8-12 year old could assemble without damaging or mis wiring it because of the tiny parts and no knowledge of what the component boards do. My 11 year old granddaughter who loves science, math and stem looked at it and walked away. I built it and then she took over running, programming and connecting to her phone. Then.....

The bad: It started falling apart. The camera module fell off, the drive motors came loose because the fasteners all unscrewed from the turning and jerking and vibration. The smallest of nuts were lost forever. My assembly issue was to overtighten and risk breaking boards or not tight enough and have them come loose.

In conclusion: Its relatively inexpensive and does work. I think I would use Locktite on screws. Also wrenches to hold tiny nuts should be included.

Function:
Allows students to **build and program robots using Arduino**.
Often includes sensors such as ultrasonic distance sensors and line followers.
Teaches coding, electronics, and robotics principles.

Product analysis



Size:

Approximately 465mm x 119mm x 284mm

Desktop-sized robotic arm suitable for laboratory or classroom work.

Aesthetic:

Industrial robotic arm design made from metal. The use of CNC'd metal gives the robot a professional appearance similar to small factory robots. Exposed mechanical joints emphasize engineering structure.

Function:

The robot is a 6-degree-of-freedom robotic arm which can be controlled via PC, mobile app, or wireless controller. Used for pick-and-place tasks and robotics programming practice. This has potential for progression as it can lead into more complex concepts such as inverse kinematics.

Cost:

Mid to high price for an educational robotics kit (around USD 150–USD 200). This high cost is due to multiple servo motors and metal construction. I believe this is unreasonable as it provides no extra educational value compared to other competitors as the user still can only control each servo independently.

Safety:

Moving points such as joints and grippers are possible spaces for children to get their fingers stuck so safety precautions such as turning off the robot must be taken. However, the low voltage prevents this from causing real harm.



Tom Carroll

★★★★★ **Quality Construction. Clear Assembly Videos**

Reviewed in the United States on March 2, 2018

Edition: LeArm | **Verified Purchase**

When I received the LeArm kit from Amazon, I was amazed at the quality of the arm, from the heavy metal base, industrial quality ball-bearing, good servos and the very clear assembly instruction videos on the Internet. It assembles easily in a few hours. I would recommend this arm over any other robot arm for robot experimenters as it will last. I intended to write about it in my monthly column in a robotics magazine and was waiting for their version of the 5-DOF hands, another manufacturer's version of which is shown in the photo. I never received those hands from LeWanSoul. I did receive their X-Arm shown to the right in the photo and found it to also be of excellent quality.

However, after a year of experimentation and demonstrations for others, I have found both of these robot arms far higher quality than any of the other robot arm kits I have tested. The LeArm uses separate 'standard' servos, each of which must be connected to a controller or Raspberry Pi or Arduino board. Depending on the position of the arm, I could manipulate almost a pound. Programming of the LeArm is fairly simple as you are dealing with standard servos that require basic pulse-width signal inputs.

The X-Arm has very similar construction but you can look at both robot arms at the Amazon site and see the physical similarities and differences in the two arms. Both use the same quality ball bearing base, though the metal structure is a bit different and the gripper/claw is identical to the other. This arm uses their LX-150 intelligent serial bus servos that can be 'daisy-chained' and the final end of the three wires of the daisy chain transfers the servo's address (each servo has a number of 1 to 6) and power through the three wires. (You can see the small connector on the end of the white cable in my photo) Your microcontroller addresses each servo independently through that cable, very similar to the Robotics Dynamixel servos. The higher cost is due to the cost of the high-cost intelligent servos.

I had a bit of trouble programming the arm with a RaspberryPi but soon got the hang of it. I ended up writing a bit of code to use it with Python, directly.

The only issue I might say is using the tiny metric screws that were all in one bag. I found some that were too long so I used washers in mounting them to the small aluminum cylinders between the servos. One of the threaded holes in the long cylinder was a few degrees off 90 degrees so I just left the screw off. The video instructions were fine though I would have preferred a written manual. I found myself 're-winding' the video to listen again 'what the girl really said.' I also would have liked to test their two hand models as theirs seems so much better made than the other company's hands shown in my photo.

I also have their 6, 16, 24 and 32 channel servo controllers that have extensively come into use with my demonstrations. The LSC-6 controller is furnished with the LeArm and is used with the hand controller (they call it a 'handle') to control the arm wirelessly. Note the receiver board in the photo of their photo of the arm.

I would highly recommend the LeWanSoul robotic arms and controllers and servo testers for students from high school to advanced university instruction, as well as home experimenters. Either of their servo models are well made with metal gears and have yet to fail for me. Their web site offers much material on their products, and they have been around long enough for one to find all sorts of articles about uses of the LeWanSoul robot arms and controllers.

→ Read less



blakebart

★★★★★ **Extremely well designed and manufactured 6DOF robotic arm with remote Bluetooth to iPhone/Android connectivity**

Reviewed in the United States on June 26, 2018

Edition: LeArm | **Verified Purchase**

I'm surprised more people haven't reviewed this robotic arm kit. I found the kit extremely well thought out. The parts machining and anodizing are just beautiful. A very important aspect of this design is the turntable that the arm sits on. It is built using a large diameter industrial quality bearing which provides a very solid base for the arm to operate on and provide servo-controlled rotation. The servos are top quality, very smooth operation and very powerful. The gripper is pre-assembled, a nice touch, and the joint design is quite clever. LeWanSoul provides excellent tutorial videos, and an extremely helpful assembly video using 3D modeling to show how things go together in sequence. Another reviewer noted the importance of sorting out all the fastener hardware, that really helped smooth the build for me. The Bluetooth connection to the free iPhone app worked great, another impressive feature of this kit, especially for the money. The PS2 controller worked fine, though I prefer the iPhone app. There is also PC interface software that is provided although I have not tried that. I'm particularly interested in connecting an Arduino to the arm and writing sketches for controlling LeArm, and Aaron at LeWanSoul showed me where to find the necessary information for that. LeWanSoul has been very responsive answering my questions. Oh, and check out the video for LeHand on the LeWanSoul website.

46 people found this helpful

Customer:

Targeted at older students, robotics hobbyists, and educators. Recommended age 14+ due to complexity and is used for learning robotics, programming, and automation.

Environment:

Durable aluminum construction increases product lifespan. However, electronic components require proper disposal when damaged. Since the product can be used for a long time, it reduces the waste caused by the product.

Product Analysis



Size:

Compact mobile robot about **90 × 180 × 130 mm** makes it well suited for classroom desks and small testing environments.

Aesthetics:

Modern educational robot design with colored metal parts and wheels. Friendly appearance suitable for classrooms and younger students. Includes a display and sensors that give the robot an interactive look.

Customer:

Designed for STEM education, schools, and coding programs and is suitable for learners aged around 12+. It is often used in robotics clubs and coding classes.

Environment:

Although it uses recyclable aluminium, it also uses ABS which can harm the environment. Since the kit uses rechargeable batteries and modular parts, nothing has to be thrown away as it can easily be recharged, repaired or upgraded.

Safety:

Rounded edges and enclosed electronics make it safer for students as it prevents them from getting cuts or from getting shocked.

Materials:

Aluminum structural parts
ABS plastic casing
Electronic sensors and PCB control boards
DC motors and wheels.

Cost:

Higher-end educational robotics kit (around USD 50–USD 250).

Higher price reflects advanced sensors, controller board, and software ecosystem.

Feature:

The kit consists of a simple 2 wheel drive which can easily be programmed in either Scratch or C++. This allows the user to not only control the robot remotely but also autonomously

SDMF1919

★★★★★ Very frustrating at first

Reviewed in the United States on December 27, 2024

Size: mBot Coding Box | [Verified Purchase](#)

It was easy to build. My 10 year old did it by himself. It took us about 4 hours to figure out how to control it though. The app in the android store is only compatible with really old phones and then once you find an old phone and get it installed, it doesn't work anyway. The chrome browser app doesn't work either. Use the QR code on the mblocks website to download and install the updated android app. Then it will finally work on your current device.

Design implications:

Over all three of the products, I have noticed 3 very distinct feature: They all have tool and none of them contain batteries. Something that I had not considered when thinking about my designs is that the user may not have the correct screwdriver. Including a screwdriver in the box would remove the extra hassle on buying one. Furthermore, the lack of batteries supports my decision of not including batteries within the package. Due to both the shipping cost and regulatory restrictions on lithium batteries and the additional cost of AA batteries for global distributor. This would defeat the original purpose of the robotics kit as a cheap and reliable resource for lightly touching on key concepts such as detailed one such as EKF. Another feature I noticed is that the nuts were are bare metal. This resulted in review complaining that the nuts would fall off. To solve this problem, I will implement nylon inserts in the nuts.



Scroll Saw:
The scroll saw has a sharp edge which quickly oscillates up and down to cut through wood

Advantages	Disadvantage
Allows for curved and intricate pattern	- Not as versatile as other saws - Can only cut a single angle

Solder:
The Soldering iron has a hot tip which is used to melt solder and solidify the connection between a wire and the pins



Advantages	Disadvantage
- Can be done in under 30 seconds - The joints can withstand forces	Creates fumes which contain lead (Depending on the solder)



Vacuum Forming:
In vacuum forming, the former heats a plastic sheet until it is soft and flexible. It then uses a vacuum to suck the plastic around a mold.

Advantages	Disadvantage
- Quick to make pieces - Simple to use	- Only useful with thermoplastics - Cannot create intricate shapes

Heat Press:
The heat press heats the textile and presses a special material onto it, forming a design on the surface.



Advantages	Disadvantage
- Easy to use - Quick to print designs - Low-Cost	Could lead to inconsistent results



Belt Sander:

A long piece of sanding paper spins around a drum, shaving off chips on the surface of the wood.

Advantages	Disadvantage
• Quick to sand parts down • Simple to use	• The surface still remains slightly rough • Cannot reach all corners of a wooden piece

Strip heater:

The wire heats up allowing it to easily cut soft materials such as styrofoam.

Advantages	Disadvantage
• Quick to make models • Good for prototypes	• Not accurate • Only works for certain materials



Pillar Drill:

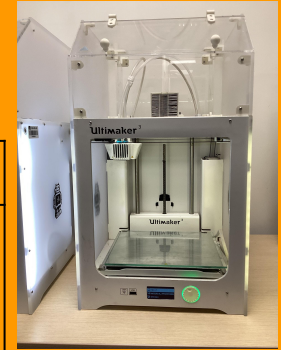
The drill bit at the top rotates very quickly to drill through a material. The lever on the right pulls the drill straight down into the part.

Advantages	Disadvantage
• Quick • Easy to drill	• Can only drill vertically • Can be unsafe in the part is not clamped

3D Printer:

The nozzle on the top heats a plastic until it becomes malleable as it lays the material down onto the print bed.

Advantages	Disadvantage
• Can make intricate pieces • Good for prototypes	• Only can make them in plastics • Slow • The print can fail





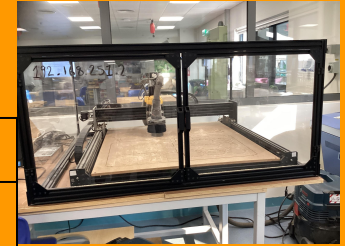
Laser Cutter:

The laser cutter burns through the wood to cuts certain pieces out of it by using a high power laser.

Advantages	Disadvantage
- Quick to make pieces - Simple to use	- Only works in 2 dimensions - Burns the material

CNC machine:

The drill bit at the top rotates very quickly. The Drill bit is moved around to remove certain parts of the material



Advantages	Disadvantage
- Good for intricate designs - Simple to use	- Slow - Only works for certain materials (depending on the drill bit) - Expensive

Design implications, From my methods of production research, I have chosen FDM 3D printing and laser cutting to manufacture the kit parts as they are relatively quick, low effort and produce results with very low tolerances. Furthermore, I will use a scroll saw to cut the packaging, the vacuum former to compartmentalize the parts and a vinyl cutter to create designs for the parts and the packaging. Finally I will use a soldering iron to create strong and permanent electrical joints.



Hand Tools:

The hand tools allow us to create designs using physical tools such as saws and handheld drills which gives us the most amount of control.

Advantages	Disadvantage
- Uses Full dexterity - No setup time - Always available	- Takes time to make - Difficult to make it

The machine quickly cuts the vinyl using a very sharp blade and moves the vinyl by moving the material vertically and the blade horizontally



Advantages	Disadvantage
- Can make intricate pieces - Good for prototypes	- Only can make them in vinyl - Only works on a relatively small scale

Material	Advantages	Disadvantages	Uses
Cardboard	• Cheap • Easy to work with • Lightweight	• Difficult to differentiate between different types • Relatively weak	• Boxes • Packaging
Plywood	• Strong and durable • Resistant to warping • Lightweight • Cheap	• Susceptible to delamination • Hard to make intricate cuts • Inconsistent quality • Sometimes contain gaps	• Roof decking • Cabinets • Wall paneling
Oak	• Dense • Aesthetically pleasing grain • Resistant to wear • Long Lifespan	• Expensive • Heavy • Can Split • Can wear tools	• Furniture • Floors • Doors • Veneers
Pine	• Cheap • Easy to cut • Light • Aesthetically pleasing grain	• Soft • Can warp • Not moisture resistant	• Furniture • Shelves • Framing • Paneling
MDF	• Smooth • No grain • Cheap • Easy to cut	• Heavy • Poor moisture resistance • Lots of dust	• Cabinet • Mouldings • Speakers • Panels

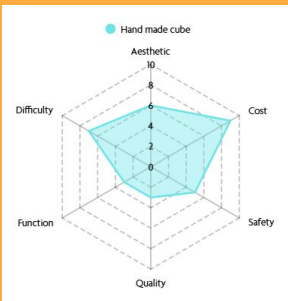
Material	Advantages	Disadvantages	Uses
Acrylic	• Clear • Light • Available in different colours • Weather resistant	• Scratches • Hard to machine • Brittle	• Signs • Screens • Lightboxes • Aquariums
PLA	• Useful for 3D printing • Biodegradable • Available in many colours	• Brittle • Not useful long term • Low strength	• Prototypes • Model Parts • Replacement Parts • Chassis • Brackets
Aluminium	• Lightweight metal • Resistant to corrosion • Easy to machine • Recyclable	• Soft • Expensive • Lower Strength	• Robot frames • Extrusions • Screws • Heat Sinks
Steel	• Strong • Cheap • Easy to machine • Long lasting	• Heavy • Rusts • Conductor • Hard to cut	• Cars • Appliances • Beams • Tools

Material Testing

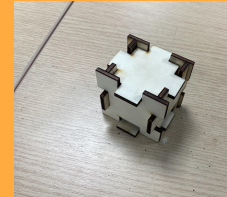
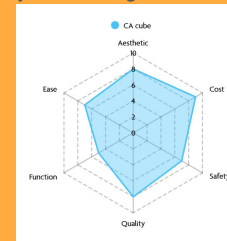
Material Name	Scratch Test	Dent Test	Drilling Test	Cutting test
Acrylic 	- Does Not scratch with a fingernail - Captured Fingerprints - Scratches with sharp objects - Scratches are only visible when shone against a light	- Brittle - Easily Dents - Quickly fractures once dented	- Quite resistant to drilling - Large amount of residue builds up around the drill hole	- Soft - Heats up quite a bit - Results in a coarse cut
MDF 	Does not scratch	Difficult to dent	Easy to drill	- Quite soft - Cuts were clean
Plywood 	- Brittle - Breaks along the grain - Easily Deforms - Scratches across the grain	- Flakes - Dents Easily	Easy to drill	- Cuts were not clean - Easy to cut

Finger Joint Cube

Making a cube with finger joints by hand was a tedious and repetitive process which did not result in a particularly good product. First, I made borders on each wood piece using the side of another wood piece. Then I marked which parts I would need to cut. 2 sides would need the all the middler sections cut out, 2 sides for need all the edge sections cut out and and the other 2 sides would need half and half cut out. To cut out the sections, I used a coping saw for coarse cuts and sandpaper for finer cuts. With the coping saw, I often cut on or beyond the marking leading to inconsistent pins and slots. As a result, I had to test each joint as many of them did not fit on the first test. Once all of the joints fit, many holes were left between each joint which I patched up and fixed in place using wood glue. Finally, I sanded everything down to get smooth surfaces and rounded edges. Overall, this process took two lessons.



Making a cube with finger joints with CAD and CAM was drastically easier than doing it by hand. To design the parts needed for the cube, I used Techsoft. Techsoft intuitive design made the process of marking the parts a whole lot easier. Once the CAD was made, I uploaded it to a USB stick as a DXF file where I plugged it into a separate computer where, using the values written on a notebook, I calibrated the power of the laser cutter. I then uploaded my design and turned on the laser cutter. Due to the age of the school's laser cutter, I had to move the laser head out of the corner before using it. Once the cut was complete, I applied some glue on the edges of each piece that would come in contact with another part. Then I sanded everything down to get smooth surfaces and rounded edges. This only took a single lesson.



Design implications: From my materials research, I have decided to use cheap materials that are easy to work with such as PLA and plywood. These materials are optimal for FDM 3D printing and laser cutting. another advantage of PLA is that it is easily shaped by the soldering iron so any incorrect measurements can easily be fixed. Furthermore, I will use HIPS (High impact polystyrene) for the interval compartmentation of the packaging and cardboard or as the exterior. HIPS is easily vacuum formed, is not harmful and very durable. This ensure none of the components will cross compartments and possibly damage each other. Cardboard is used as the exterior packaging as it is often very strong due to its corrugated midsection. Furthermore, it is easy to fold making and is very cheap to get. The fasteners will be made of aluminum as aluminium screw as readily available in my area and are strong enough to thread themselves into the PLA without requiring heat-press inserts.

Size

The robot kit must be able to fit into a small box of around 300mm x 200mm 50mm with the final robot being able to easily fit on a small desk. The individual parts must be able to fix in the user's hand for easier manipulation of elements. Hence the final bot will be around 200mm x 200mm x 200mm.

Function

Problem research::

The kit must be able to accept some sort of input from a single or multiple devices and output a resulting motion or action. After completing the kit, the user must understand both basic and relatively advanced concepts of robotics

Safety

Safety Standards my product will have to pass:
CPSIA
CE Marking
IEC 62368-1



The BSI (British Standards Institution) is the national body which regulates technical and Safety standards across the country. It also is the body which test products for their safety and reliability and certifies them as legally sellable. They do regular tests on each product to ensure that the product still remains reliable and safe. When designing my product, I will make sure that it follows the BSI standard such as protection against electric shocks by using suitable insulation, fire hazards by using safe voltages, mechanical risks and hazards from environmental conditions use. From this, my product will receive the BSI Kitemark and this will ensure the consumer that the product is safe to use and marketable in England

Components

A cheap microprocessor will be used as the brains of the whole robot as it is cheap and readily available. Alongside that, I will use gold MG946R servos because of their all metal gear which prevent gear skipping along with a TT motor which permits ample speed without being dangerous.

The CPSIA (Consumer Product Safety Improvement Act) ensure that the product does not include any or limited harmful chemicals such as lead and Phthalates to protect children. It also mandates that the product has to go through a form of third party testing to ensure it does not harm the child.

The CE mark on products states that the product is allowed to be sold in the EU (European Union) and EEA (European Economic Area). For a product to be legally sold in the EU and the UK it must pass the EU's safety, health and environmental standards. My product must always abide by these standards as to permit sale within Europe which (as stated in my in-depth research) is an extremely large market.

Design Implications:

To keep my product safe in the hands of children I will employ a number of safety features which can be separate into 2 categories: electrical and mechanical. To prevent the children from getting shocked, I will use a battery with less than 12V, Fuses to prevent overcurrents, use proper insulation on wires, clearly label the power input and place the batteries in a safe holder. Mechanically, I will fillet the edges, make sure that the screws cannot fall out by using either threaded inserts or nuts to secure the parts, use large parts to prevent young children from choking and use safe and non-toxic materials such as PLA and cardboard.

Skills it will teach: The kit will teach the use about basic concepts such as gears, racks and pinion and with motors along with programming skills such as sensor control, PIDF loops. These allow the user to not only build their mechanical skills but also their coding skills

Name		Description	Application	Advantage	Disadvantage
Spray Paint		Spray painting is a type of painting where the paint is atomised before being released into the air and landing on the part	- Shake the spray can - Start spraying the paint while slowly moving over the part from side to side	- Fast - Smooth finish - Reaches difficult to reach places - Less physical effort: - Versatility:	- Wastes a large amount of paint - Requires preparation - Not safe - Expensive
Varnish		A clear, protective coating made of resin, solvent, and a drying oil that provides a hard, lustrous finish to surfaces like wood	- Dip the brush into the white varnish - Varnish the part by brushing along the grain - Wait 5 min before doing another coat	- Protection - Durability - Appearance - Ease of Application	- Slow Drying - Yellowing - Difficult Cleanup - Bubbles could form - Multiple coats - Strong Odour
Wax		Wax is a protective coating made from natural or synthetic waxes that enhances the wood's natural grain, and offers moisture resistance	- Use a piece of cloth to rub the wax - Rub the part with the cloth	- Aesthetic - Easy To Apply - Environmentally Friendly	- Weak - Requires maintenance - Difficult to remove - Wears off quickly
Stain		Stain is a coloured substance which 'bleeds' into wood, changing its colour while still showing the grain	- Dip the brush into the stain - Stain the part by brushing along the grain - Wait 5 min before doing another coat	- Aesthetic - Protection against moisture, mildew and sun - Easy to apply - Cheap	- Has to be replaced - Does Not hide imperfections - Few colours
Paint		Paint is a coloured substance which form a decorative or protective layer over a part	- Dip the brush into the paint - Paint the part by brushing along the grain	- Aesthetic - Durable - Easy to clean - Versatile - Eco-friendly - Cheap	- Difficult to remove - Requires prep
Lacquer		Lacquer is a clear protective layer which provides a glossy finish	Spray the part vertically	- Fast Drying - Durable - Aesthetic	- Toxic - Yellowing

Design Implications: From this research, I have chosen to spray paint my parts as a primary background colour and to hand paint or use a stencil to draw designs onto the parts. I chose spray painting to paint large areas for its ease of use and its variety of colours. I have chosen to hand paint for intricate details as spray painting is too broad for small designs. However, I may use spray painting through a stencil for the designs if I am incapable of painting myself. I chose these painting styles over using coloured filament as there is only a select few colours to choose from and wastes a large amount of plastic when switching colors. Although I claimed to mainly use PLA for my product, I continued to research woods as it enables me variety if I am to use laser cutting to manufacture parts and woods are often better to apply a finish upon. From my aesthetics research on page 7, I have decided to use some form of primary blue and use primary red as accents.

Wood

Fixing and construction methods

Metal



Finger Joint:

A finger joint cuts the wood in interlocking wedges, increasing surface area to create a stronger adhesive bond, spread out stresses and is versatile

Dovetail joint:

Instead of cutting straight wedges, flared tails are cut out of the wood which intersect to create a strong joint. This design makes them less susceptible to lateral forces, they are more aesthetic than the finger joint and are more durable.



Butt joint,

a butt joint is the simplest type of wood joint making it quick to do but makes the joint weak and it may be difficult to perfectly align

Adhesive:

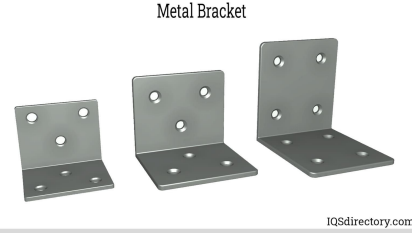
An adhesive is a substance which connects 2 materials together. This allows 2 different materials to be joined together differently and distributing stress and are easy to apply



Design Implication: Since I am mainly using PLA as my material and the kit must be able to be assembled disassembled and reassembled, Nuts and bolts with brackets would be ideal because they form strong connections which can be easily broken down. Though heat press inserts can be used instead of nuts, they require are often made of brass which is more expensive than aluminium. As a robotics kit, attempting to minimise cost is one of the most important qualities of the kit so using heat inserts would be wasting unnecessary money.

Metal Bracket:

Metal brackets are essential structural fasteners used to connect, support, and reinforce materials, commonly made from steel, aluminum, or iron. There are many types of metal bracket: L, Z, T, Flat.

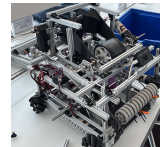


Welding:

Welding works by melting base metals to physically bind the 2 pieces of metal together. This results in an extremely strong joint. However, the equipment required for this process is expensive

Rivet:

Rivets are pieces of metal which pass through holes within the metal. The end of the rivet is then deformed, often by a hammer, creating a second head on the other side of the metal pieces. This creates a strong bond between the pieces of metal



Nuts and bolts:

Bolts are screws with a constant width. They are passed through a hole and on the other side of the pieces of metal, a nut is screw and secured tightly onto the metal

Ergonomics

Overview:

The ergonomics and anthropometrics of my product are important as they dictate how comfortable it is to use, controlling how likely they are to recommend it to others. More specifically, anthropometrics is the study of measuring human beings while ergonomics is how those measurements can be implemented into designs to make it more comfortable to use. To do this, I will take measurements from my client and use this anthropometric data to design a produce which is somewhat easy to build, understand and learn from. I need not only take into account o the robot physically feels when building it but also how loud it is when running, how understandable the instructions/videos are.



Percentile	5th	50th	95th	client	Source(other than client)
Hand length (mm)	155	172	189	175	PMC7720326
Hand breath (mm)	63	74	85	80	PMC7720326
Grip diameter (mm)	32	42	52	45	Springer BMC Pediatrics 2021
Hand grip strength (N)	98	178	290	130	Acta Paediatrica 2002
Finger thickness (mm)	12	15	19	16	ScienceDirect 2005/2010

Cognitive ergonomics:

This aspect of the product refers to how the product is perceived. This includes details such as the clarity of the instructions, how the code is presented and how concepts are explained. As an educational kit, this will be one of the most important aspects as without clarity, the box would be a bunch of parts.

Physical Ergonomics:

This is the aspect of the product which the user will physically interact with. Hence, the product must not consist of any sharp corners, the surfaces must be smooth. To keep the surfaces smooth, I will sand the surfaces using gradually increasing grits of sandpaper and use countersunk nuts and bolts to maintain this smoothness throughout the robot. It should also be easy to transport thus be light. The recommended carry weight for a teenager is approximately 10% of their body weight. Since the minimum age of the target audience is 13 y.o (whose average body weight is 45kg) the robot should weigh less than 4.5kg. In addition, the robot should also not be extremely loud. This is an important aspect of the robot as often electronic parts can make lots of noise such as motor brush (the noise which motor brushes make while they run) which produces between 45 dB and 60 dB of noise which is significantly above the largely accepted volume of background noise as 35 dB according to the BS 8233:2014.Reducing this will make the kit hugely more bearable.

From my research, I have concluded that I must ensure that all holes must be bigger than 20mm (1mm added tolerance) to permit the user to easily access the screw holes using their fingers. Additionally, All parts other than fasteners must be larger than a 2 cm in any dimension. This ensures that the user will be able to hold the part with at least two fingers and stabilize it without falling. Furthermore, I will make sure that it weighs less than 2 kilogram. The makes sure that the user is able to easily pick up and move the parts and the full robot around with ease. The kit will also include both an instruction manual and a QR code including all of the instruction to build the kit and explain each part of it. The packaging will be a cuboid with curved bottom edges to enable easy stacky and maximum space utilisation when shipping and is comfortable to hold from the bottom.

Area	Specific Point	User Requirements	9.1: It must be a programmable, cheap and slightly difficult to assemble as to encourage the user to utilise their own critical thinking to solve problems on their own and should take more than 2 hours to assemble physically. 9.2: The product will include both an instruction manual and a QR for the instructions for ease of assembly.
Aesthetics	1.1: My product should look like a unique robot but based on something real in order to intrigue the audience for which this product is targeted for i.e it should make 75% o pupils interested in the product as determined by a post-build survey of at least 10 individuals in the target audience.. 1.2: The colours of the product will be bright and vibrant and must consist of mainly primary colours to encourage creativity and problem solving.	Maintenance	10.1: The product must require minimal actions to be taken to keeps its functional value as once its built, the user should have understood all of the concepts. Once this has happened the product has completed its mission.
Packaging	2.1: The budget will be £30 as it is a main priority of my product is that it is cheap to reach as many students as possible while maintaining a safety margin if any of the designs include extra components and to compensate for the volatility of electronics prices.. 2.2: I want to price my product at £40 as it is at price low enough to be affordable but still high enough to make the product high quality and is mentioned by my client as a reasonable price (£50 ■ £43).	Ergonomics	11.1: The corners will be chamfered and the screws will be easily accessible so that everyone, no matter their hand size, can build it..
Client	3.1 My client is Faris Tarraf as he already has a keen interest in robotics however, is unable to participate in competitions due to his location. 3.2: My wider target market is children and teenagers as they are often building their interests during this time and are often the main target for large competitions.. 3.3: Ideally, my product would be sold on amazon and would work with school to possibly integrate it into the curriculum or as a CCA as it allows the product to reach the widest target audience.	Manufacturing	12.1: Most of the product will be 3d printed or laser cut and secured together with Metal nuts and bolts. I may include some wooden panels as well. This allows for more modularity of the robot and the undoing of steps.
Environment	4.1: My product will try to reduce the amount of negative effects it will have on the environment by using eco-friendly materials for at least 60% of the mass of the robot as this will make consumers more likely to buy the product. 4.2: Due to the use of electronics, they product as a whole will not be recyclable. However, the use of PLA permits the main body to be recycled and the electrons to be reused..	Product Lifespan	13.1 The product should last about a year at the least under normal temperature and humidity conditions and of daily play. 13.2: Once used, the 3d printed body can be thrown away and the arduino could be used to make other projects.
Size	5.1: When built, the product will be around 20cm x 20cm by 20cm so it can be held in the hand and can be worked on easily as my client has a small workspace..	Packaging	14.1: The product will require compartmentalised packaging to maximise space efficiency and to prevent parts from damaging each other. 14.2: The packaging will be a cuboid with a rounded bottom edge for ergonomic and comfortable handling and warehouse space efficiency.
Safety	6.1:The final product will have secure wire connections, protected electronics and double insulation. This ensures that it is able to get the CE and Kitemark stickers. Furthermore, it will not have sharp corners and will use less than 12V, be		
Function	7.1: The kit will be able to educate the user on basic and engineering and robotics concepts such as different types of gears and lightly touching on and applying advanced concepts such as CPG gaits and Extended Kalman Filters.		
Materials	8.1: The body will be made of PLA and the electronic will be outsourced,		

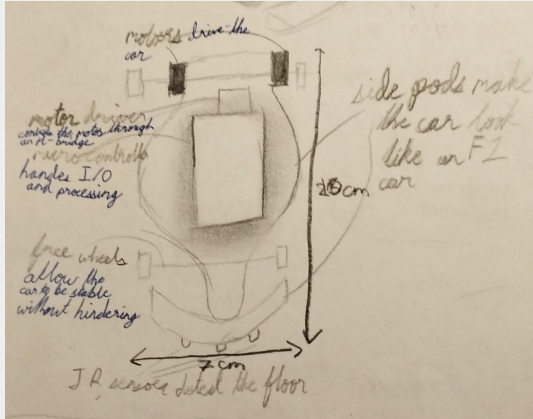
Kit for a line following robot

Pros:

- Cheap to manufacture
- Teaches basic sensors
- Easy to assemble
- Low part count
- Lightweight
- Visually engaging

Cons:

- Limited progression
- Limited customisation



Teaches basic sensor and motor control and internal gears

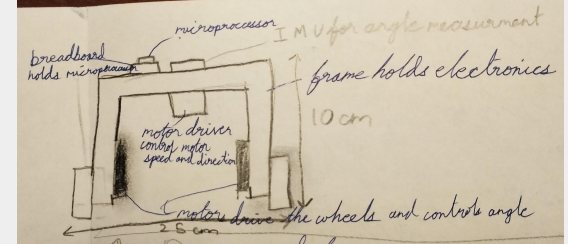
Kit for a 2-wheel balancing robot

Pros:

- Introduces PIDF control
- Teaches basic sensors and data protocols
- Compact
- Low material usage

Cons:

- Difficult to tune
- Rectangular appearance makes the robot less engaging



Teaches basic sensor and motor control different protocols and PIDF

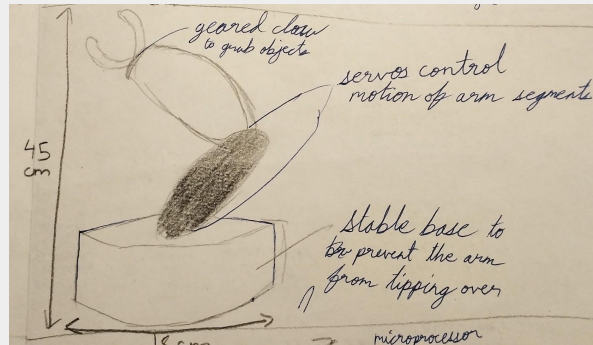
Kit for a robotic arm

Pros:

- Teaches Inverse Kinematics
- Highly interactive
- Easy to demonstrate
- Have good mechanical aspects

Cons:

- Servo wiring may be damaged if the arms moves beyond limits.
- Low portability
- High cost



Teaches motor control and inverse kinematics

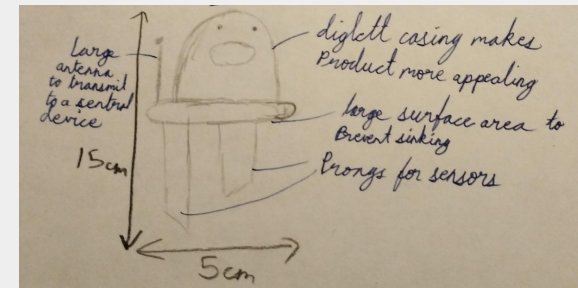
Kit for an Environmental Monitoring robot

Pros:

- Strong STEM links
- Used in the real world
- Easy to expand

Cons:

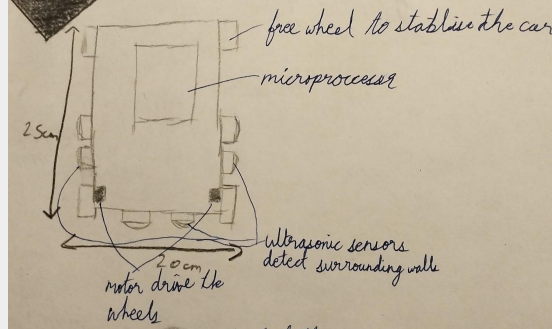
- Less movement
- Limited mechanical content



Teaches wireless connectivity, scientific method, how sensors work, different data protocols

Kit for a maze solver robot

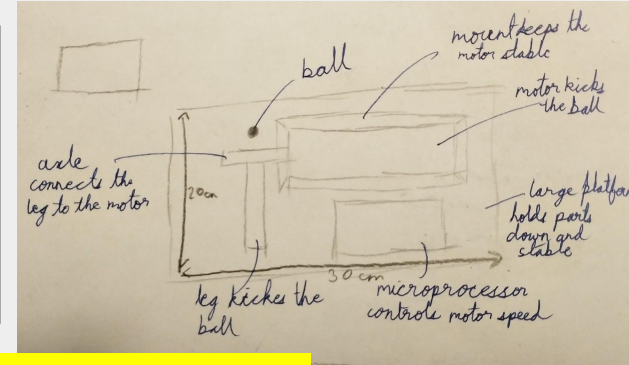
Pros:
Teaches search algorithms
Teaches sensors
Introduction to Micromouse
Can be progressed
Cons:
Difficult to debug
Requires a maze to fully test
Limited mechanical complexity



Teaches sensor and motor control, search algorithms

Football kicker

Pros:
Fun and engaging
Teaches motion systems
Strong visual appeal
Cons:
Educational value lower
Limited expansion
Difficult to justify curriculum use

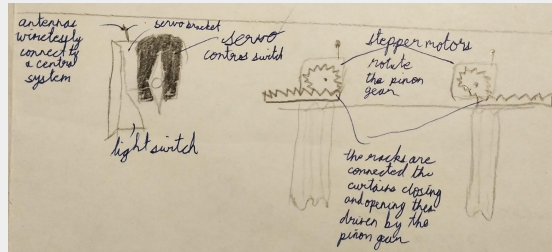


Teaches basic motor control

Kit for automatic light switch and curtains

Pros:
Real world application
Teaches automation
Relatively low cost

Cons:
Less engaging
Less Robotics focused
Limited mobility

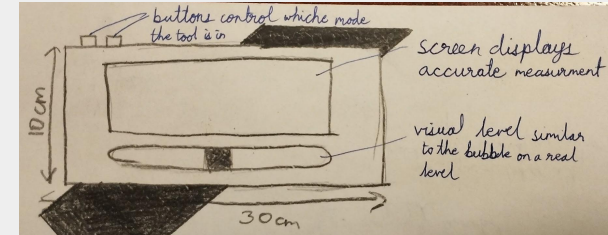


Teaches sensor and motor control, embedded systems

Kit for an electronic leveler

Pros:
Teaches protocols and Sensors
Real life application

Cons:
Not visually appealing
Limited progression
Lacks locomotion

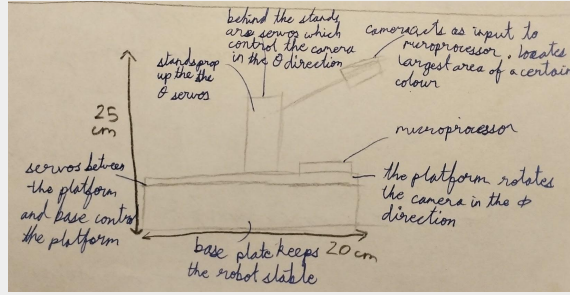


Teaches sensor control, different forms of output, LED and OLED

Kit for a colour tracking arm

Pros:
Teaches computer vision
Visually impressive
Modular
Programming focus

Cons:
High cost
Stationary base

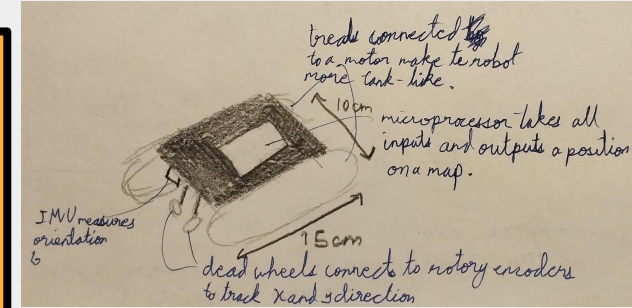


Teaches Inverse kinematics, computer vision processing, motor control

Kit for a tank drive robot with odometry

Pros:
Teaches navigation
Teaches Sensors
Expandable into FTC
Real world applications

Cons:
Higher Cost
Limited mechanical complexity

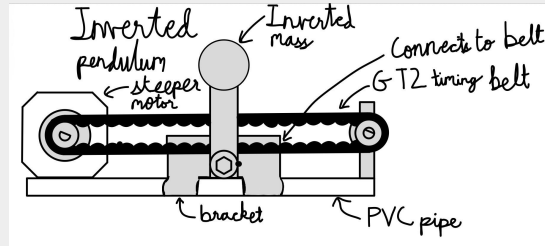


Teaches Forward kinematics, basic drivetrain systems, odometry

Kit for a inverted pendulum

Pros:
Teaches deep control theory
Teaches advanced mathematics

Cons:
Difficult to introduce
Easily damaged
Very Difficult to tune

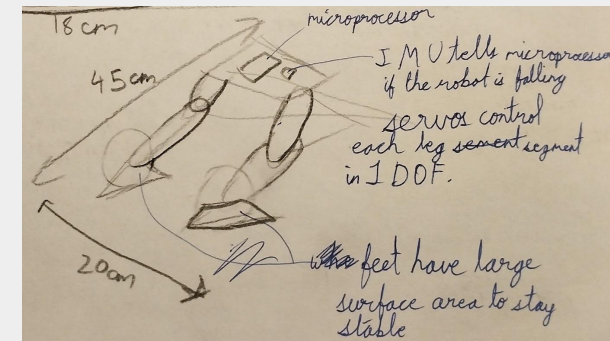


Teaches control theory, sensor and motor control, physics of pendulum

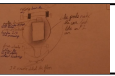
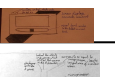
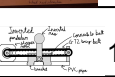
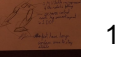
Kit for a bipedal robot

Pros:
High education potential
Teaches movement
Teaches inverse kinematics
Teaches balance
Expandable
Unique

Cons:
High cost
Long build time
Easy to fail



Teaches, Sensor and motor control, inverse kinematics, PIDF, control theory

		Aesth													
Idea	1	Cost 2	Client 3	Enviro 4	Size 5	Safety 6	Func 7	Matt 8	User 9	Maint. 10	Ergo 11	Manu. 12	Life. 13	Total	
	1	5	5	4	4	2	5	2	5	3	5	4	5	4	53
	2	4	5	4	4	3	4	3	5	5	4	4	5	5	55
	3	4	3	4	4	3	4	4	5	3	3	4	3	3	47
	4	5	4	4	4	4	5	2	3	2	4	4	4	4	49
	5	3	5	3	4	5	4	3	4	3	4	3	4	4	49
	6	3	5	3	4	5	4	3	4	3	4	3	4	4	49
	7	3	4	2	3	2	3	3	2	3	5	3	2	5	40
	8	5	4	2	4	2	4	2	3	2	5	3	5	5	46
	9	5	3	5	3	5	3	4	4	4	4	4	4	3	51
	10	4	5	4	3	5	5	4	3	3	4	3	2	4	49
	11	3	3	3	4	4	3	5	4	4	3	3	3	4	46
	12	5	3	4	4	5	4	5	4	5	3	4	5	5	56

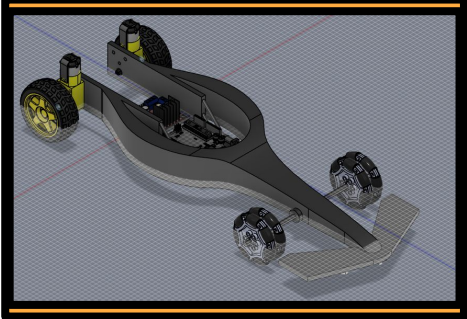
The scores were awarded between 1 and 5 through subjective thought between me and my client. Alongside, we reached out to other family members or friends on certain scores. The orange rows are the designs which will be developed designs. This matrix is a general scoring comparing to the specification and does not go in-depth into sub topics as the main purpose of this ,matix is to narrow the 12 initials ideas into 4 developed ideas. Any score lower than a 3 would require a major redesign to fix.

Design implications: Since Designs 1,2,9 and 12 had the 4 highest ratings, they will move onto the proposed design section.

Line following robot

The line following robot is a robot which is able to follow a black line against a white background. It uses 2 rear motors to drive the car and 2 sets of omni wheels to allow the car freedom of movement when turning. It uses a set of 3 IR sensors to detect the darkness of the ground below it. It was modeled after an F1 car and students can use it to learn about basic motor usage and how different sensors work.

The parts will come separated but would only need to be pushed together. The plastic bushings would be pushed into the nose of the robot along with a pair of omni wheels on each side. In the middle, the arduino and the motor driver will be mounted. At the back lie the motors and wheels



TT motors drive the back wheels

The side pods make the car look like an F1 car

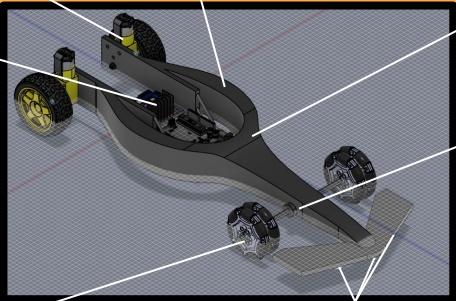
The main body is PLA as it is easily 3D printable and relatively strong

The L298N motor drivers and arduino are situated in the middle to make the physics of the car more similar to F1 cars

Bushings reduce the friction between the axle and the hole.

Dual set of omni wheel ensures that at least 1 side-roller is in contact with the ground, preventing increased wheel deterioration

3 IR sensors at the bottom detect changes in the colour of the floor which informs the arduino to turn.



I would 3D print the plates using PLA. The screws, wheels and the motor will be store bought.

"I quite like this design. It is nice how it was structured around an F1 car. This one is beautiful. It does look a bit simplistic so it wouldn't teach me as much but would offer a good entry into the field of robotics" - Faris

Pros	Cons
Cheap to produce	Takes a long time to print
Easily customisable	Omni wheels require tight tolerances
Lightweight	PLA is brittle
No fasteners required	The design is too simple to be able to be carried further

1.1	1.2	2.1	2.2	3.1	3.2	3.3	4.1	4.2	5.1	6.1	7.1	8.1	8.2	9.1	9.2	10.1	11.1	12.1	13.1
Green	Green	Green	Green	Purple	Yellow	Purple	Green	Red	Red	Yellow	Red	Green	Red	Red	Red	Green	Green	Red	Green

Green - fully meets spec
Yellow - partially meets spec
Red - Doesn't meet spec
Purple - not applicable

2 Wheel balancing bot

The dual wheel balancing robot is a robot which is able to self align itself upright simply through acceleration of the bottom two wheels. This will be done through the collaboration between an Inertial measurement unit and the arduino. Despise the hardware, the user will have to learn about PIDF and feedback loops to stabilize the machine. This design also allows for more further development of the mechanism as it can be adapted for other uses such as a remote controlled car or an object movement system.

The main chassis will be made of PLA as it is soft enough for threads to catch onto and is easy to 3D print cheaply.

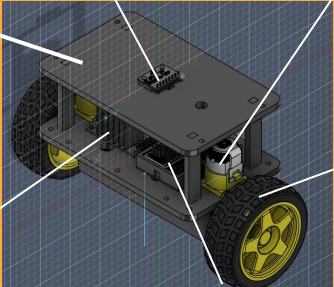
MPU 6050 measure the rotational speed which can be converted into an absolute angle.

TT motor drive the body of the robot up by applying lateral motion different to the motion of the CoM

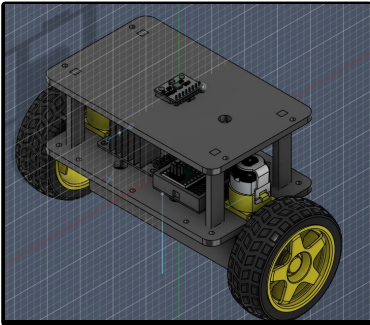
The wheel tread will be made of rubber or TPU to increase friction and stability with the floor

The L298N motor driver controls the motors direction and speed

Using an arduino nano as the micro controller as it is powerful enough for this application and its small form factor makes it ideal for this size



Assembly:
First, the cover on the breadboard would be removed and stuck into the recession on the surface of the plate. Then, the L298N and the 2x TT motors would be bolted onto the base plate. Once bolted, the standoff would be attached along with the top baser plate and MPU 6050. Finally the wheels would be attached onto the motors



I would 3D print the plates using PLA. The screws, wheels and the motor will be store bought. These will all go into a cardboard box on which I spray paint over a vinyl stencil. The related parts will be packaged together in sections made by the vacuum former.

“The robot is very simplistic and appears to easily teach the basic concepts such as basic motor control, basic velocity algorithms and Sensor inputs and protocol such as UART and I2C.the aesthetic of this design is functional but lacks visual appeal for the target user group, which may reduce engagement.” - Faris

Pros	Cons
Cheap to manufacture	Structure might break because of the tin standoffs
Low print time	Flexible PLA causes noise ito IMU readings
Teaches about algorithms such as PIDF	

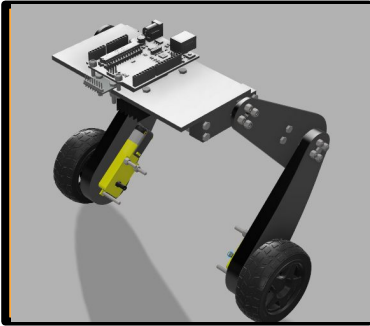
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Yellow	Green	Yellow	Green	Purple	Red	Purple	Yellow	Yellow	Red	Green	Red	Green	Green	Yellow	Green	Green	Green	Green	Green

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Yellow - partially meets spec
Red - Doesn't meet spec
Purple - not applicable

Wheeled Bipedal Robot

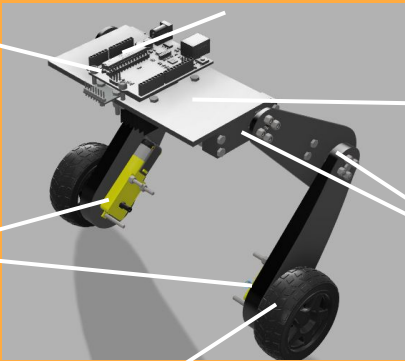
The concept for this prototype is to act as a racing robot through its freedom of movement. Due to its strong base motors, the robot will be able to sprint along straight but the individual vertical freedom on each leg allows the robot to tilt into corners. Not only this, the robot will be able to ride across terrains without being tipped over. It is through this freedom from which we can teach the users about feedback loops such as PIDF and a light introduction into kalman filters.

The user would first attach all of the servos to servo brackets. Two of these would then screw into the base plate. Then the second pair of servo brackets would mount onto the upper leg. Then, servo horns are screwed into the brackets and attached to their respective servo. Then the arduino is bolted onto the plate



An arduino uno is used as the microcontroller because it accepts a standard barrel jack connector with USB-B connectivity while maintaining an acceptable clock speed and RAM size

The MPU 6050 is used as an IMU for the robot.



Main chassis will be made of PLA as it is soft enough to thread into, hard enough to support electronics and easy to prototype upon

TT motors drive the robot to help it stabilize itself when still. They also help lock the wheel to increase friction when the robot walks

Tread will be manufactured out of rubber to maximise friction between the robot and the floor.

MG946R servo motors control the height of the robot and allow it to pitch up and down and roll right to left

The chassis will be made out of 3D printed PLA as it is easy to prototype and soft and secure to thread. The fasteners, motors and wheels will be store bought.

"The bipedal robot looks similar to the balancing bot but more complex. The extra servo joints make the product seem a lot more difficult which I am up for but may deter an audience who wants to get into robotics. It's also very ugly." - Faris

Pros	Cons
Complex mechanism permit deep understanding	High print time
Soft PLA for easy threading	Screws may get stripped if used repeatedly
Free leg geometry	Takes a long time to build
	Most difficult design

1.1	1.2	2.1	2.2	3.1	3.2	3.3	4.1	4.2	5.1	6.1	7.1	8.1	8.2	9.1	9.2	10.1	11.1	12.1	13.1
Green	Green	Red	Yellow	Purple	Green	Purple	Yellow	Yellow	Green	Yellow	Green	Yellow	Green	Green	Yellow	Green	Red	Green	Green

Green - fully meets spec
Yellow - partially meets spec
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Purple - not applicable

Colour tracking turret

The colour tracking turret is able to autonomously use a camera connected to an arduino to detect a certain colour and follow that colour around using a main servo controlling ϕ and a secondary servo controlling θ in a 3d polar coordinate system.

The main body will be made of PLA

MG946R gold shaft servo motors control the axis ϕ and θ of the camera

A Heavy base is used to prevent the motor from moving the base instead of the dome



Camera searches for the largest area of a certain colour in its view

All electronics will be bolted onto the platform to prevent parts from moving when they don't have to.

The whole dome rests on a lazy susan bearing. This allows the whole dome to rotate instead of just the camera

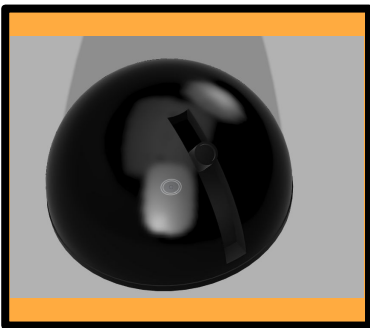
The lazy susan bearing will be made out of aluminium to present any deformations under load that may affect turning.

The kit will contain numerous parts: 4x hoods segments, 4x base segment, 2x servos, 2x servo brackets, 1x arduino uno, 1x battery case 1x axle and 1x camera.

The user would start off sliding the hood segments together. Then they would attach the servos to the servo brackets. Once attached, 1 of the servo would be bolted through the bracket and into the base while the other is bolted onto of that one. Then, the camera would be attached to the axle and the axle would be attached to the second servo. This leaves the user with a physical model which will become functional.

The overarching shell will be made with 20% infill PLA while the base will be made of 100% infill PLA. The lazy susan bearing will be store bought as it is made out of metal which is difficult to scale to larger audience when operating as a single person.

“This looks like you smashed together a missile launcher from an american warship and put Star Wars together or a planetary. Either way, I really like the turret design. However, it is a bit too static how it doesn't really move around like the other designs but overall it is a good design.” - Faris



Pros	Cons
Easy to 3D print	Cannot track fast moving objects
Easy integration of lazy susan bearing	Lazy susan must be bought
PLA is suitable for static parts	PLA may warp
Low part count	Curved surface is prone to bad aesthetics

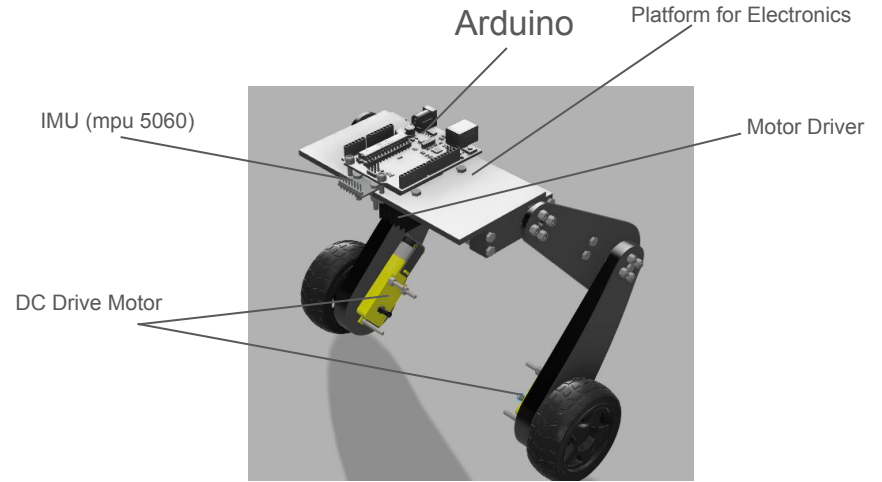
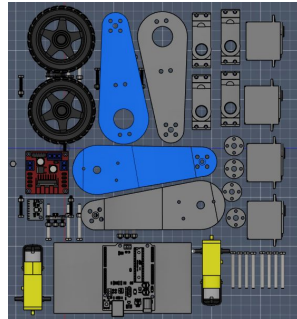
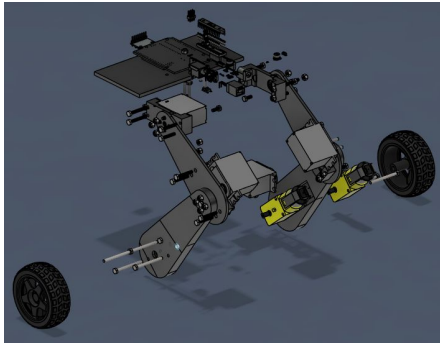
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Yellow	Green	Red	Yellow	Purple	Green	Purple	Yellow	Yellow	Green	Yellow	Yellow	Green	Green	Green	Red	Red	Green	Green	Green

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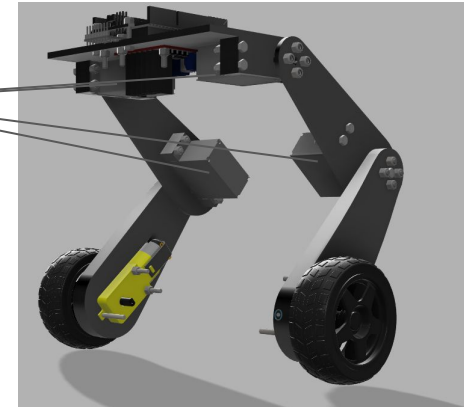
Proposed Solution

I chose the wheeled biped concept design as it contains the best track for transitioning between basic and advanced robotics. It provide basic functionality with individual servo legs to advanced functionality through Bipedal CPG gaits which replicate Regular bipedal locomotion. It does hold the downside of being slightly more expensive than the other proposed solutions but I believe that the added difficulty improves the effectiveness of the design. Compared to the balance bot, it teaches a wider range of mechanical concepts. Compared to the colour tracker, it offers greater physical and programming challenge. Compared to the line follower, it provides a deeper progression path.

The concept for this prototype is to act as a racing robot through its freedom of movement. Due to its strong base motors, the robot will be able to sprint along straight but the individual vertical freedom on each leg allows the robot to tilt into corners. Not only this, the robot will be able to ride across terrains without being tipped over. It is through this freedom from which we can teach the users about feedback loops such as PIDF and the link between maths and engineering through inverse kinematics

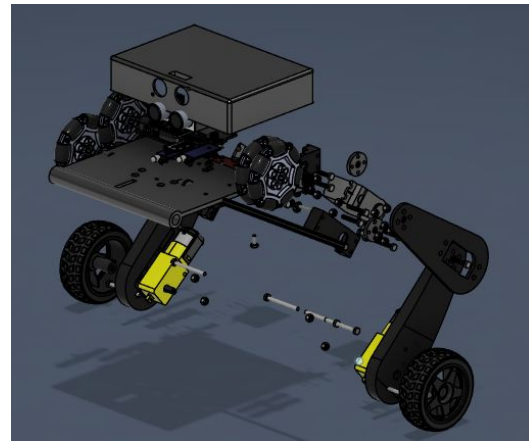


Servo leg piece control

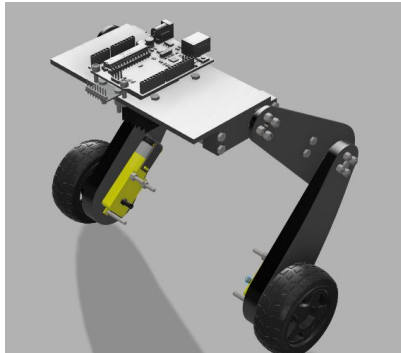


Changes from Proposed Design into V1

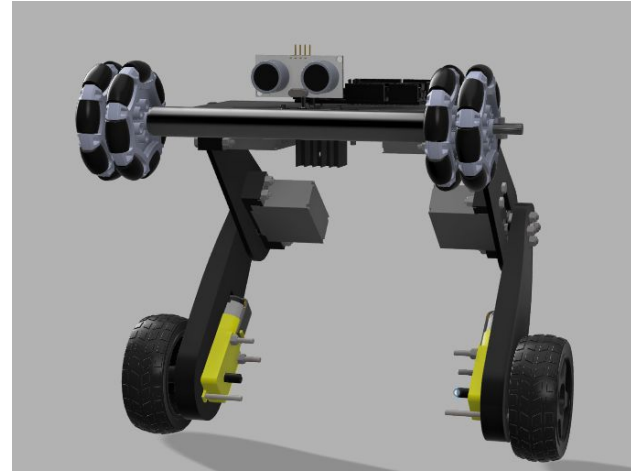
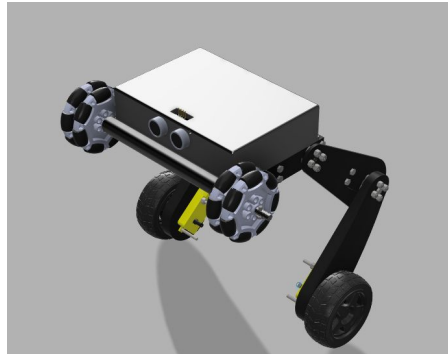
- After client feedback,
- I had added a protective cover around the arduino and electronics.
- I have also removed all sharp corners from all parts to prevent unnecessary discomfort during building.
- I also rearranged the location of electronics for a better weight distribution.



Before



After



Developed V1 Physical Prototype

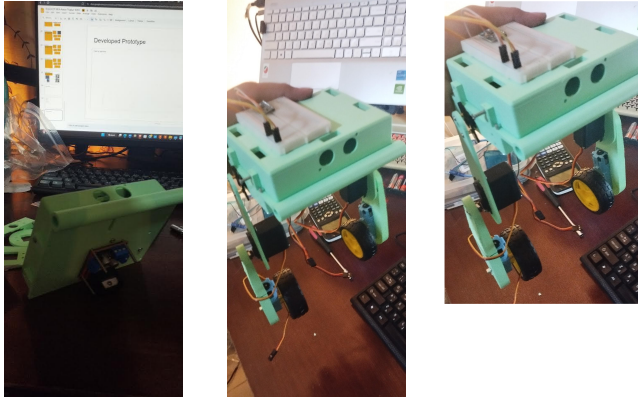


Fig 1, 2, 3:
Complete
assembly of
V1



Fig 4: All
3D printed
parts
separately



Fig 5: Original
bracket design

Throughout the building process, I continuously analysed the building experience and how I could improve the process. I was able to identify many flaws within my design.

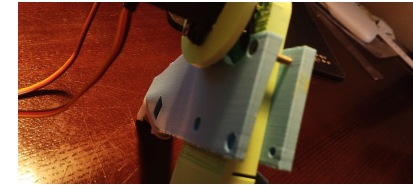
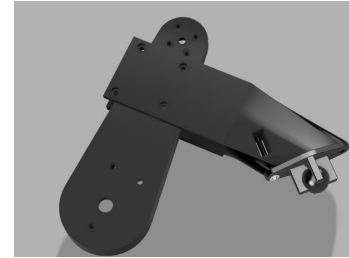
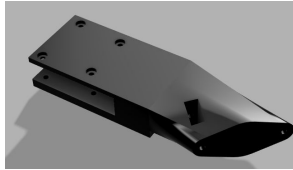
- Often when designing, I needed the robot to stand up straight. This often proved difficult due to the bipedal nature.
- Although I had designed for the wires to funnel through a single hole, it proved difficult to do as wires often disconnected from the pins on the Arduino. This may be because the PCB was previously used but it still presents as a possible problem.
- Each time I screwed something into the part, the material would often sink into the hole and the screw would extend above the surface, interfering with Electronics and sometimes even shorting the pins on the Arduino Uno.
- Furthermore, the motor brackets were often too small for the motor themselves. When 3D printing, the filament has a given thickness which interferes with part tolerances. Even though the slicer on most 3D printers accounts for this inconsistency when making the G-code, it can still present as a problem when dealing with cheaper 3D printers.
- This also caused issues with the fitting of the wheel have me to resort to adding it of the inside of the robot instead of the outside.
- It also became obvious once I started building that I did not require 2 of the servo brackets (more specifically the lower 2) as the hole for the servo was slightly too small for the servo and the servo shaft would not extend beyond the part if the bracket was included.
- The Breadboard was originally designed to be placed at the top but it makes the design look more confusing and makes wiring unnecessarily difficult.
- I had expected lateral sag within the legs as both PLA and the servo horn are flexible but the degree to which the parts flex was of concerning matter.
- It would also help if the cover was hinge as it would not need to be removed and bolted back in every time that I want to make a change.

Design Implications:

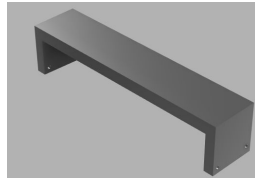
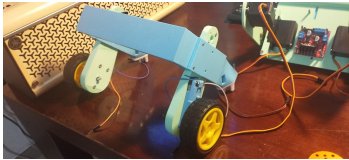
From that experience, I have decided to split the design into 2 more steps: a V1.5 and a V2. The V1.5 will include additional modules to provide basic functionality to the prototype to test electrical circuits and code. The V2 will contain all of the stated improvements of this build analysis.

Physical Prototype V1.5

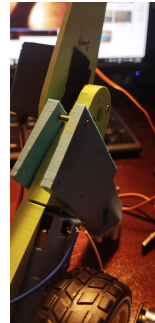
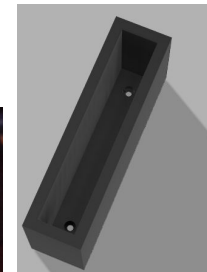
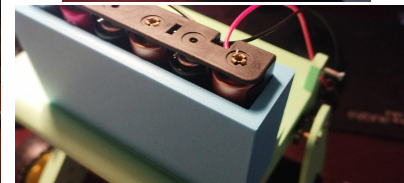
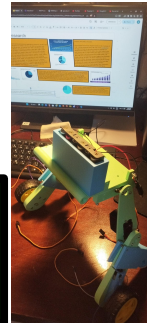
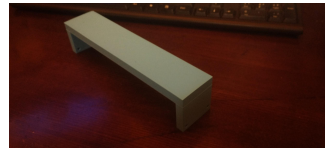
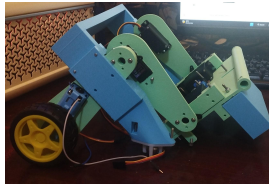
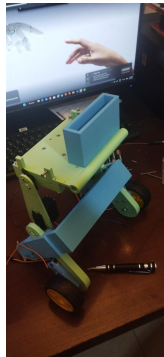
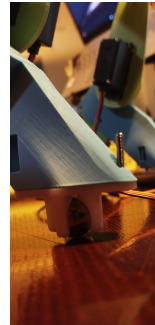
To continue to wiring and coding V1, I would require some extra pieces. As noted in my build analysis, I would require a slot to fit the batteries into. Due to voltage restrictions on the arduino uno, I would have to connect the battery pack to 2 separate sources. Furthermore, to test basic functionality I also printed 2 supporting braces on the lower legs with caster wheels on the ends. This allows the robot to easily stand upright and permits me to test basic drive functionality. Again mentioned in my build analysis, the lateral tolerance of the legs is unacceptable. To prevent a major redesign of the legs, I have opted to use a cross brace connecting the two supporting brackets.



While attaching these modules onto the design, I noticed some further issues. Since the battery holder would require bolts, The main base plate would have to include holes for these. As a temporary solution, I used a soldering iron to burn a hole through the PLA. Additionally, the main cover for the electronics would not fit, hence continued with this model without the cover and one of the screws for the servo bracket won't thread fully through. These are issues to be fixed in V2. Furthermore, For a quick and easy solution on the mounting of the support brackets, I have used screws to clamp onto the leg. This will be reiterated in V2 where I will include a dedicated screw hole for the part.



The red circle shows where the PLA seeped out when using the soldering iron to make holes.



Design Implications: As a summary of the changes I will make, I will add screw holes to the base plate, I will redesign the battery holder and add the ability to attach support wheels. I will redesign the legs as to reduce lateral flex and result in inconsistent 'foot' placement.